Status	Finished
Started	Tuesday, 29 September 2026, 3:49 PM
Completed	Tuesday, 29 September 2026, 3:54 PM
Duration	4 mins 36 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. order()
- c. arrange()
- d. sort()

The correct answer is: sorted()

Question **2**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Merging
- b. Sorting
- c. Searching
- d. Complexity

The correct answer is:
Complexity

Question **3**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Sorting the entire list in one pass
c. Dividing the list into halves
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting is rarely used in programming
c. Sorting helps in finding duplicates quickly
d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Merge Sort
- b. Binary Search
- c. Bubble Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are discarded
- b. They are split again into smaller sublists
- c. They are combined to form a single sorted list
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It complicates the selection of items
- b. It makes the data unsorted
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Removing duplicates from a list
b. Finding a specific element in a list
c. Merging two datasets
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It complicates the analysis process
c. It makes the analysis slower
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is less efficient for large lists
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort is difficult to implement
- d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **13**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Sorting data sequentially
- c. Sorting data in a single pass
- d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **14**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It divides the list into sublists
- c. It uses the heap data structure
- d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **15**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. Duplicates
c. Searching
d. All of the above

The correct answer is: All of the above